Suppose we have used the Comparison Test to show that a series $\sum a_n$ converges by comparison with a series $\sum b_n$. If we approximate the series $\sum_{k=1}^{\infty} a_k$ by the nth partial sum $\sum_{k=1}^{n} a_k$, then the remainder is

$$R_n = s - s_n = \sum_{k=n+1}^{\infty} a_k.$$

Similarly, let

$$T_n = \sum_{k=n+1}^{\infty} b_k$$

be the remainder of the approximation of the series $\sum_{k=1}^{\infty} b_k$ by the nth partial sum $\sum_{k=1}^{n} b_k$. Since $a_n \leq b_n$ for all n, we have $R_n \leq T_n$. If $\sum b_n$ is a p-series $\sum \frac{1}{n^p}$, then, as we have seen, we can estimate its remainder T_n as

$$T_n = \int_n^{\infty} \frac{1}{x^p} dx.$$

If $\sum b_n$ is a geometric series, then T_n is the sum of a geometric series, which we can sum exactly. In either case, we know that R_n is smaller than T_n.

1. Show that $T_n = \dfrac{ar^n}{1-r}$ when $\sum b_n$ is the geometric series $\sum_{n=1}^{\infty} ar^{n-1}$.

2. Use the sum of the first 100 terms to approximate the sum of the series $\sum \frac{1}{n^3+1}$. Estimate the error involved in this approximation.

3. Use the sum of the first 10 terms to approximate the sum of the series $\sum_{n=0}^{\infty} 5^{-n}\cos^2(n)$.